(ii) The given second integral can be written as

$$\int \frac{px+q}{\sqrt{ax^2+bx+c}}dx = \int \frac{A(2ax+b)+B}{\sqrt{ax^2+bx+c}}dx$$

$$= A\int \frac{2ax+b}{\sqrt{ax^2+bx+c}}dx + B\int \frac{1}{\sqrt{ax^2+bx+c}}dx$$

(The first integral is of the form $\int f'(x)[f(x)]^n dx$)

$$= A\left(2\sqrt{ax^2+bx+c}\right) + B\int \frac{1}{\sqrt{ax^2+bx+c}}dx$$

The second term on the right hand side can be evaluated using the previous types.

Examples 11.40

Evaluate the following integrals

(i) $\int \frac{3x+5}{x^2+4x+7}dx$ (ii) $\int \frac{x+1}{x^2-3x+1}dx$ (iii) $\int \frac{2x+3}{\sqrt{x^2+x+1}}dx$ (iv) $\int \frac{5x-7}{\sqrt{3x-x^2+2}}dx$

Solution

(i) Let $I = \int \frac{3x+5}{x^2+4x+7}dx$

$$3x+5 = A\frac{d}{dx}(x^2+4x+7)+B$$

$$3x+5 = A(2x+4)+B$$

Comparing the coefficients of like terms, we get

$$2A = 3 \Rightarrow A = \frac{3}{2}; 4A+B = 5 \Rightarrow B = -1$$

$$I = \int \frac{\frac{3}{2}(2x+4)-1}{x^2+4x+7}dx$$

$$I = \frac{3}{2}\int \frac{2x+4}{x^2+4x+7}dx - \int \frac{1}{x^2+4x+7}dx$$

$$= \frac{3}{2}\log\left|x^2+4x+7\right| - \int \frac{1}{(x+2)^2+\left(\sqrt{3}\right)^2}dx$$

$$= \frac{3}{2}\log\left|x^2+4x+7\right| - \frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{x+2}{\sqrt{3}}\right)+c$$

(ii) Let $I = \int \frac{x+1}{x^2-3x+1}dx$

$$x+1 = A\frac{d}{dx}(x^2-3x+1)+B$$

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